Program

Master's, Doctoral and Postdoctoral Track Program: Computer and Data Science

1. Open Doors winner's skill set

The winner should know:

- key concepts, theoretical foundations, and scope of computer science and mathematical analysis;
- major research paradigms and methodological approaches in technologies related to computer and data science, including machine learning, artificial intelligence, and data analysis;
- current trends, challenges, and emerging directions in computer architecture, networking, and operating systems;
- software verification methods and project management;
- information security regulations;
- cryptographic techniques;
- modern programming languages, integrated development environments (IDEs), and version control systems.

The winner should be able to:

- apply theoretical knowledge of Software Engineering to solve practical and research problems;
- develop, code, debug, test, and optimize algorithms and programs;
- perform OS administration, software deployment, and network service configuration.

2. List of degree programs

2.1. List of doctoral programs

1.2.1 Artificial intelligence and machine learning

1.2.2 Mathematical modelling, numerical methods and software packages

2.3.1 System analysis, management and information processing, statistics

2.3.2 Computing systems and their elements

2.3.5 Mathematical and software foundation for data science

2.2. List of master's degree programs

01.04.02 Applied mathematics and informatics

02.04.02 Fundamental Informatics and information technologies

09.04.01 Informatics and computer engineering

Program

09.04.02 Information systems and technology

09.04.03 Applied informatics

09.04.04 Software Engineering

3. Content

Field of science 1. Applied mathematics

Mathematical analysis

1. Limits of sequences and functions. Continuity of functions of one and several variables. Differential calculus of functions of one and several variables. Derivative and differential, extrema of functions of one and several variables
2. Integral calculus of a function of one variable. Indefinite integral. Definite integral over an interval. Integral with variable upper limit. Improper integrals
3. Functional series. Power series. Fourier series and Fourier integral

Field of science 2. Computer science, information systems

Computer science

1. The concept of information: general characteristics of data collection, transmission, processing, and storage; information measurement; units of information; means and methods of data transmission
2. Information systems and information technologies: concepts and definitions. Architecture design of information systems. Information support
3. Number systems. Conversion of numbers from one number system to another. Arithmetic operations in positional number systems
4. Data representation in digital computers. Data types. Integers, floating-point numbers, characters, and strings. Simple and composite data types
5. Database design, development, administration, and optimization. SQL
6. Algorithms and algorithm design. Algorithm flowcharts. Algorithmic problems
7. Theoretical and practical foundations of programming languages. Fundamentals of procedural and object-oriented programming

Field of science 3. Computer science, artificial intelligence

Artificial intelligence systems

1. Concepts and terminology of AI, Knowledge, Data, Big Data, Artificial Intelligence
2. Methods of data analysis. Data visualization, Descriptive statistics, correlation analysis, regression analysis, anomaly and outlier search. The main stages of data mining

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3. Machine learning. The main types of machine learning tasks. Machine learning algorithms. Ensemble methods of machine learning. Evaluating the effectiveness of machine learning models. Quality metrics
4. Artificial neural networks. An artificial neuron. The architecture of the ANN. Principles of learning. Generative ANN. Problems of creating and using an ANN
5. Applied artificial intelligence systems. Computer vision, Natural language processing, Expert systems

Field of science 4. Computer science, interdisciplinary applications

Computer science

1. The concept of information: general characteristics of data collection, transmission, processing, and storage; information measurement; units of information; means and methods of data transmission
2. Information systems and information technologies: concepts and definitions. Architecture design of information systems. Information support
3. Database design, development, administration, and optimization. SQL

Field of science 5. Computer science, software engineering

Software Engineering

1. Software: definition, properties. Software engineering: purpose, basic principles and concepts. Systems engineering. Business reengineering. Defining of roles in the project team
2. Software architecture. Architectural modeling. The software lifecycle, models, and development processes
3. Software requirements development. Requirements management. Types, properties, and formalization of requirements. The requirements work cycle. Organization of software requirements development
4. Software development project management. Goals and processes of feasibility analysis for software projects. Key resources for ensuring the software lifecycle

Field of science 6. Computer science, hardware & architecture

Computer architecture, networks and systems

1. The concept of a computer system. Basic criteria for classification of computer systems. Michael Flynn's taxonomy
2. The concept of computer architecture. Major types of computer architecture. The hardware and software concept. Hardware support for operating-system functions. Approaches to enhancing computer performance
3. The concept of an operating system (OS). Operating-system architecture. Operating-system modes. Types of operating-system kernels. Features of the microkernel OS architecture
4. Functional components of an OS. Process management in the OS. Memory management in the OS. Application programming interface and methods of its implementation

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5. File systems (FS) as part of a computer system. Mountable file systems. Methods of disk space allocation in file systems. Major file systems for personal computers
6. The concept of a computer network. Purpose, composition, and classification of computer networks. The OSI network model. Data transmission methods and security in computer networks. Components of a structured cabling system. IPv4 addressing

4. Preparation materials

4.1. Recommended reading

Field of science 1. Applied mathematics

Zorich, V. A. Mathematical analysis I (2nd ed., R. Cooke & O. Paniagua T., Trans.). 2015 URL: https://ndl.ethernet.edu.et/bitstream/123456789/51201/1/Vladimir%20A.%20Zorich.pdf
Zhang, T. Mathematical analysis of machine learning algorithms : [электрон. ресурс] / T. Zhang. Cambridge : Cambridge University Press, 2023. URL: https://tongzhang-ml.org/lt-book/lt-book.pdf
Pugh, C. C. Real mathematical analysis : учебник / C. C. Pugh. — 2nd ed. — (Undergraduate texts in mathematics). — Cham : Springer, 2015 URL: https://smartmanmaths.wordpress.com/wp-content/uploads/2017/11/real-mathematical-analysis.pdf

Field of science 2. Computer science, information systems

Tanenbaum A.S. et al. Computer networks. Prentice-Hall International Editions , 1996. URL: https://theswissbay.ch/pdf/Gentoomen%20Library/Networking/Prentice%20Hall%20-%20Computer%20Networks%20Tanenbaum%204ed.pdf
Hopcroft J. E., Motwani R., Ullman J. D. Introduction to automata theory, languages, and computation. 2006 URL: https://www-2.dc.uba.ar/staff/becher/Hopcroft-Motwani-Ullman-2001.pdf
Rainer R.K., Prince B., Cegielski C. G. Introduction to Information Systems. John Wiley & Sons Singapore Pte. Limited, 2015. URL: https://humdiana.files.wordpress.com/2018/03/introduction-to-information-system-edisi-5-tahun-2014.pdf

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Field of science 3. Computer science, artificial intelligence

Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. MIT Press, 2016. URL: https://www.deeplearningbook.org/
Russell S., Norvig P. Artificial Intelligence: A Modern Approach. Publisher: Pearson, 2021 URL: http://lib.ysu.am/disciplines_bk/efdd4d1d4c2087fe1cbe03d9ced67f34.pdf
Alpaydin E. Introduction to Machine Learning. London: The MIT Press, 2010 URL: https://kkpatel7.files.wordpress.com/2015/04/alppaydin_machinelearning_2010.pdf

Field of science 4. Computer science, interdisciplinary applications

Ferguson N., Schneier B., Kohno T. Cryptography Engineering: Design Principles and Practical Applications 1st Edition, Wiley, 2011 URL: https://www.schneier.com/wp-content/uploads/2015/12/fortuna.pdf
Isaaks E.H., Srivastava R.M. An introduction to applied geostatistics. Oxford University Press, Oxford, 1989 URL: https://www.geokniga.org/bookfiles/geokniga-anintroductiontoappliedgeostatistics.pdf
Sayama, H. Introduction to the modeling and analysis of complex systems : [электрон. ресурс] : учебник / H. Sayama. — Milne : Open SUNY Textbooks, 2015. URL: https://knightscholar.geneseo.edu/oer-ost/14/

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Field of science 5. Computer science, software engineering

Kim, G., Humble, J., Debois, P., & Willis, J. The DevOps handbook: How to create world-class agility, reliability, & security in technology organizations, 2016
URL: https://malenezi.github.io/malenezi/SE489/Books/The%20DevOps%20Handbook%20-%20How%20to%20Create%20World-Class%20Agility,%20Reliability,%20and%20Security%20in%20Technology%20Organizations.pdf

Wiegers, K. E. Software requirements / K. E. Wiegers, J. Beatty. — 3rd ed. — [S. l.] : Microsoft Press, 2021.
URL: https://liteka.ru/english/library/7358-software-requirements#1

Pacheco D., Sgro S. Principles of Software Architecture Modernization: Delivering engineering excellence with the art of fixing microservices. BPB Publications, 2024
URL: https://oceanofpdf.com/authors/diego-pacheco/pdf-epub-principles-of-software-architecture-modernization-delivering-engineering-excellence-with-the-art-of-fixing-microservices-monoliths-and-distributed-monoliths-english-edition-download/

Field of science 6. Computer science, hardware & architecture

David A. Patterson , John L. Hennessy. Computer Organization and Design RISC-V Edition. Elsevier Science, 2005
URL: https://vowi.fsinf.at/images/9/95/TU_Wien-Rechnerstrukturen_VO_(Grünbacher)_-_David_A._Patterson_John_L._Hennessy_Computer_Organization_and_Design_BookFi.org_-2.pdf

Harris D., Harris S. Digital Design and Computer Architecture. Publisher: Morgan Kaufmann, 2012
URL: https://www.r-5.org/files/books/computers/hw-layers/hardware/digital-desigh/David_Harris_Sarah_Harris-Digital_Design_and_Computer_Architecture-EN.pdf

Tanenbaum A., Austin T. Structured Computer Organization. Publisher: Pearson, 2012
URL: https://dl.rubens.cloud/books/Structured%20Computer%20Organization%20%286th%20Edition%29%20by%20Andrew%20S.%20Tanenbaum%20and%20Todd%20Austin.pdf

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4.2. Recommended online courses

Field of science 1. Applied mathematics

Online course	Link	Course description
Calculus: Single Variable Part 1 - Functions	https://www.coursera.org/learn/single-variable-calculus	The course provides the necessary theoretical material on the use of Taylor series, Lopital's rule, and Big O notation.
Calculus: Single Variable Part 2 - Differentiation	https://www.coursera.org/learn/differentiation-calculus	Course covers derivatives, differentiation rules, linearization, higher derivatives, optimization, differentials, and differentiation operators
Calculus: Single Variable Part 3 - Integration	https://www.coursera.org/learn/integration-calculus	Course covers integrating differential equations, techniques of integration, the fundamental theorem of integral calculus, and difficult integrals

Field of science 2. Computer science, information systems

Online course	Link	Course description
Information Systems Specialization	https://www.coursera.org/specializations/information-systems	The course block covers the modeling of information systems to prepare for development, discusses the capabilities of corporate systems and management aspects of their selection and implementation, the choice of IT infrastructure and change management.
Introduction to Computer Science and Programming	https://www.coursera.org/specializations/introduction-computer-science-programming	The course presents the basic principles of computing and mathematical foundations, basic programming skills for interactive and graphical applications, and numerical mathematics tools for problem solving and modeling.
Introduction to Computer Information Systems Specialization	https://www.coursera.org/specializations/introduction-computer-infosystems	The course provides information about computer hardware, software, and data security.

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Field of science 3. Computer science, artificial intelligence

Online course	Link	Course description
Data Science Project Based Learning	https://www.geeksforgeeks.org/courses/data-science-live	The course provides the basics of statistics, mathematical analysis, optimization, and statistical concepts used in ML. Topics of data analysis, SQL, databases, machine learning are considered.
Deep Learning	https://www.coursera.org/specializations/deep-learning	The course covers the topics of creating and learning neural network architecture, such as convolutional neural networks, recurrent neural networks, and transformers.
Fundamentals of Machine Learning and Artificial Intelligence	https://www.coursera.org/learn/fundamentals-of-machine-learning-and-artificial-intelligence	This course introduces students to the basics of machine learning (ML) and artificial intelligence (AI), and examines the relationship between artificial intelligence, ML, deep learning, and the emerging field of generative artificial intelligence (AI).

Field of science 4. Computer science, interdisciplinary applications

Online course	Link	Course description
IT Security: Defense against the digital dark arts	https://www.coursera.org/learn/it-security	This course covers the basic concepts and tools of IT security, including threats, attacks, and data encryption, as well as network security solutions and how to create a layered security architecture.
UCSanDiegoX: Computer Graphics	https://www.edx.org/learn/computer-graphics/the-university-of-california-san-diego-computer-graphics	The course focuses on three key competencies: creating 3D computer graphics, developing programs for viewing scenes in real time, and creating photorealistic images using ray tracing techniques.
Geographic Information Systems (GIS)	https://www.coursera.org/specializations/gis	The course covers geospatial analysis and cartographic visualization issues applicable in various sectors, including agriculture, urban planning, and healthcare.

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Field of science 5. Computer science, software engineering

Online course	Link	Course description
Software Development Lifecycle	https://www.coursera.org/specializations/software-development-lifecycle	The course aims to explore the issues of creating secure software using SDLC methodologies and to compare different methodologies based on different constraints.
IBM DevOps and Software Engineering Professional Certificate	https://www.coursera.org/professional-certificates/devops-and-software-engineering	The course familiarizes students with the basic practices, tools and technologies of DevOps, learn about working with Python, Linux shell scripting, projects on GitHub, Docker, Kubernetes, CI/CD and cloud technologies.
IBM Full Stack Software Developer Professional	https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer	The course covers full-stack development, cloud technologies, and artificial intelligence tools for building, deploying, testing, and managing applications using technologies such as Node.js, React, Docker, Kubernetes, and others.

Field of science 6. Computer science, hardware & architecture

Online course	Link	Course description
Computer Architecture	https://www.coursera.org/learn/comparch	This course introduces the fundamental concepts of computer architecture, focusing on the design and organization of computer systems. Key topics: processor design, memory hierarchy, and input/output systems.
Introduction to Hardware and Operating Systems	https://www.coursera.org/learn/introduction-to-hardware-and-operating-systems	The course covers basic knowledge about hardware and operating systems. It provides topics on computer components, interfaces, and peripherals, as well as IT tasks such as setting up a workstation and troubleshooting.
Hardware and Operating System Foundations Specialization	https://www.coursera.org/specializations/hardware-and-operating-system-foundations	The course provides an overview of the functions, management, and security of Windows and Linux operating systems, command-line tools, registry management, and system security protocols.